QUICK GUIDE

PULSE WAVE

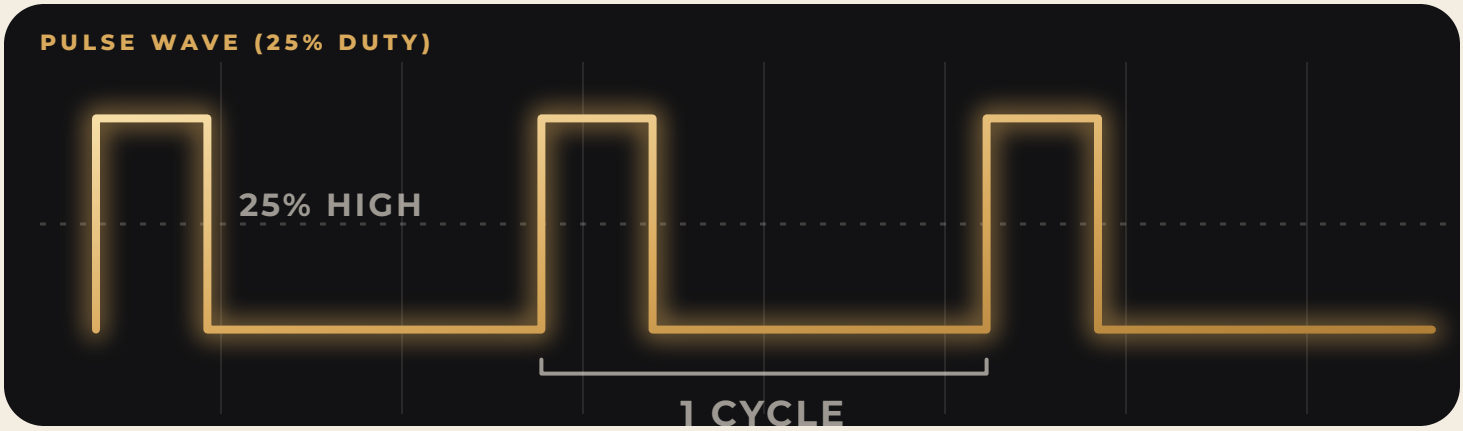
What it is, how to spot it and what to do with it.

WHAT IT IS

A pulse wave is a square wave with an adjustable width. The width, called the duty cycle, is the share of each cycle the wave spends high.

HOW IT SOUNDS

Full at 50%. Thinner and more nasal as the pulse gets narrower.



01

DUTY CYCLE

The share of each cycle the wave stays high.

02

50% IS A SQUARE WAVE

Narrower pulses sound thinner and more nasal.

03

PULSE WIDTH MODULATION

Sweep the width and the sound moves.

DO THIS TONIGHT

Sweep the pulse width slowly and listen to the tone change.

- [ ] Open any synth with a pulse width control.
- [ ] Sweep it slowly from 50% down to 10%.
- [ ] Hear the tone go from full to thin.



PRACTICE & CHECK

PULSE WAVE: YOUR TURN

WHERE YOU'LL MEET IT AS A DJ

- Classic synth bass and lead sounds.
- Pulse width modulation gives a moving, chorus-like tone.
- Chiptune and retro game music.

QUICK CHECK

1. What is a 50% pulse wave?

2. What does a narrower pulse sound like?

3. What does duty cycle measure?

Answer key: 1) A square wave. 2) Thinner and more nasal. 3) How long the wave stays high in each cycle.

REMEMBER

Change the width, change the tone.

YOUR NOTES

UP NEXT White & Pink Noise

